

Course: DevOps & Cloud Engineering
Module: Linux Fundamentals – Day 6

Ans: stsystemctl command

Ans)first: `sudo apt install openssh-server -y`

```

bravekha@LAPTOP-CK10PC9R:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
TriggeredBy: ● ssh.socket
             Docs: man:sshd(8)
                  man:sshd_config(5)
lines 1-6/6 (END) ...skipping...
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
TriggeredBy: ● ssh.socket
             Docs: man:sshd(8)
                  man:sshd_config(5)

```

Ans)systemctl status cron

```

root@kali:~# systemctl status cron
● cron.service - Regular background program processing daemon
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-10-06 11:09:13 UTC; 12min ago
     Docs: man:cron(8)
    Main PID: 113 (cron)
      Tasks: 1 (limit: 4472)
     Memory: 476K (peak: 2.3M)
        CPU: 28ms
    CGroup: /system.slice/cron.service
            └─113 /usr/sbin/cron -f -P

Oct 06 11:09:13 LAPTOP-CKI0PC9R systemd[1]: Started cron.service - Regular background program processing daemon.
Oct 06 11:09:13 LAPTOP-CKI0PC9R cron[113]: cron.service: Referenced but unset environment variable evaluates to an empty string: EXTRA_OPTS
Oct 06 11:09:13 LAPTOP-CKI0PC9R cron[113]: (CRON) INFO (pidfile fd = 3)
Oct 06 11:09:13 LAPTOP-CKI0PC9R cron[113]: (CRON) INFO (Running @reboot jobs)
Oct 06 11:17:01 LAPTOP-CKI0PC9R cron[601]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 06 11:17:01 LAPTOP-CKI0PC9R cron[603]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
Oct 06 11:17:01 LAPTOP-CKI0PC9R cron[601]: pam_unix(cron:session): session closed for user root

```

4. Check the status of the Docker service.

Ans) systemctl status docker

```
sreelekha@LAPTOP-CKI0PC9R:~$ systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-10-06 11:25:16 UTC; 39s ago
     Invocation: 94257e6e02db4559bd90eb22fe889119
   TriggeredBy: ● docker.socket
      Docs: https://docs.docker.com
    Main PID: 1599 (dockerd)
       Tasks: 19
      Memory: 29.5M (peak: 31.4M)
         CPU: 317ms
      CGroup: /system.slice/docker.service
              └─1599 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.229894258Z" level=info msg="Restoring containers: start."
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.250570958Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.282429359Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.511316859Z" level=info msg="Loading containers: done."
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.517590259Z" level=info msg="Docker daemon" commit=29.1.3-0ubuntu4.1 containerd-snapshotter=true st
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.518043159Z" level=info msg="Initializing buildkit"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.544041759Z" level=info msg="Completed buildkit initialization"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.551777460Z" level=info msg="Daemon has completed initialization"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.551898060Z" level=info msg="API listen on /run/docker.sock"
Oct 06 11:25:16 LAPTOP-CKI0PC9R systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-23/23 (END)
```

5. Verify whether the SSH service is enabled to start automatically during system boot.

Ans) systemctl is-enabled ssh

```
sreelekha@LAPTOP-CKI0PC9R:~$ systemctl is-enabled ssh
disabled
sreelekha@LAPTOP-CKI0PC9R:~$
```

- enabled → starts automatically
- disabled → does not start automatically

6. Display all dependencies required by the Nginx service.

Ans) systemctl list-dependencies nginx

```
sreelekha@LAPTOP-CKI0PC9R:~$ systemctl list-dependencies nginx
nginx.service
sreelekha@LAPTOP-CKI0PC9R:~$
```

7. View the logs generated by the SSH service.

Ans)journalctl -u ssh

```
sreelekha@LAPTOP-CKI0PC9R:~$ journalctl -u ssh
-- No entries --
sreelekha@LAPTOP-CKI0PC9R:~$
```

8. Display the last 20 log entries from the system journal.

Ans)journal -n 20

```

root@kali:~# journalctl -n 20
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.540403752Z" level=info msg="Completed buildkit initialization"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.551774602Z" level=info msg="API listen on /run/docker.sock"
Oct 06 11:25:16 LAPTOP-CKI0PC9R system[1]: Reloading finished in 175 ms.
Oct 06 11:25:16 LAPTOP-CKI0PC9R system[1]: Reloading requested from client PID 1794 ('systemctl') (unit init.scope)...
Oct 06 11:25:16 LAPTOP-CKI0PC9R system[1]: Reloading...
Oct 06 11:25:16 LAPTOP-CKI0PC9R system[1]: Reloading finished in 175 ms.
Oct 06 11:25:17 LAPTOP-CKI0PC9R system[1]: Starting ubuntu-fan.service - Ubuntu FAN network setup...
Oct 06 11:25:17 LAPTOP-CKI0PC9R system[1]: Finished ubuntu-fan.service - Ubuntu FAN network setup.
Oct 06 11:25:18 LAPTOP-CKI0PC9R dbus-daemon[133]: [system] Reloaded configuration
Oct 06 11:25:18 LAPTOP-CKI0PC9R dbus-daemon[133]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='packagekit.service' requested by ':1.25'
Oct 06 11:25:18 LAPTOP-CKI0PC9R system[1]: Starting packagekit.service - PackageKit Daemon...
Oct 06 11:25:18 LAPTOP-CKI0PC9R PackageKit[1871]: daemon start
Oct 06 11:25:18 LAPTOP-CKI0PC9R dbus-daemon[133]: [system] Successfully activated service 'org.freedesktop.PackageKit'
Oct 06 11:25:18 LAPTOP-CKI0PC9R system[1]: Started packagekit.service - PackageKit Daemon.
Oct 06 11:25:18 LAPTOP-CKI0PC9R sudo[181]: pam_unix(sudo:session): session closed for user root
Oct 06 11:26:11 LAPTOP-CKI0PC9R kernel: mini_init (119): drop_caches: 1
Oct 06 11:30:23 LAPTOP-CKI0PC9R PackageKit[1871]: daemon quit
Oct 06 11:30:23 LAPTOP-CKI0PC9R system[1]: packagekit.service: Deactivated successfully.

```

9. Display all logs generated today.

Ans) journal --since today

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ journalctl --since today  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: Linux version 6.18.33.2-microsoft-standard-WSL2 (root@f1bbfb02316b) (gcc (GCC) 13.2.0, GNU ld (GNU Binutils) 2.41) #1 SMP PREEMPT_DS  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: Command line: initrd=initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=8 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: KERNEL supported cpus:  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: Intel GenuineIntel  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: AMD AuthenticAMD  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: BIOS-provided physical RAM map:  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: BIOS-e820: [mem 0x00000000000e0000-0x00000000000e0fff] reserved  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: BIOS-e820: [mem 0x0000000001000000-0x00000000001fffff] ACPI data  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: BIOS-e820: [mem 0x0000000002000000-0x000000000f4bffff] usable  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: NX (Execute Disable) protection: active  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: APIC: Static calls initialized  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: DMI not present or invalid.  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: Hypervisor detected: Microsoft Hyper-V  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: Hyper-V: privilege flags low 0xae7f, high 0x3b8030, ext 0x42, hints 0xa4e24, misc 0xe4bed7b6  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: Hyper-V: Nested features: 0x3e0101  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: Hyper-V: LAPIC Timer Frequency: 0xc3500  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: Hyper-V: Using hypercall for remote TLB flush  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: clocksource: hyperv_clocksource.tsc_page: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: clocksource: hyperv_clocksource.ms: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: tsc: Marking TSC unstable due to running on Hyper-V  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: tsc: Detected 2495.998 MHz processor  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: e820: update [mem 0x00000000-0x00000fff] usable ==> reserved  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: e820: remove [mem 0x000a0000-0x000fffff] usable  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: last_pfn = 0xf4c00 max_arch_pfn = 0x40000000  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: x86/PAT: Configuration [0-7]: WB WC UC- WB WP UC- WT  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: Using 68 pages for direct mapping  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: RAMDISK: [mem 0x04beb000-0x04e9ffff]  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: ACPI: Early table checksum verification disabled  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: ACPI: RSDP 0x000000000000E000 000024 (v02 VIRTUAL)  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: ACPI: XSDT 0x000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: ACPI: FACP 0x000000000101000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: ACPI: DSDT 0x000000000101100 01E104 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: ACPI: FACS 0x000000000101114 000040 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: ACPI: OEM0 0x000000000101154 000064 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: ACPI: SRAT 0x00000000011F33C 000030 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: ACPI: APIC 0x00000000011F66C 000088 (v04 VIRTUAL MICROSOFT 00000001 MSFT 00000001)  
Oct 06 11:09:13 LAPTOP-CKI0PC9R kernel: ACPI: Reserving FACP table memory at [mem 0x101000-0x101113]
```

10. Monitor system logs in real time.

Ans) journalctl -f

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ journalctl -f  
Oct 06 11:25:18 LAPTOP-CKI0PC9R dbus-daemon[133]: [system] Reloaded configuration  
Oct 06 11:25:18 LAPTOP-CKI0PC9R dbus-daemon[133]: [system] Activating via systemd: service name='org.freedesktop.PackageKit' unit='packagekit.service' requested by ':1.25'  
(uid=0 pid=1867 comm="/usr/bin/gdbus call --system --dest org.freedesktop PackageKit label=kernel")  
Oct 06 11:25:18 LAPTOP-CKI0PC9R systemd[1]: Starting packagekit.service - PackageKit Daemon...  
Oct 06 11:25:18 LAPTOP-CKI0PC9R PackageKit[1871]: daemon start  
Oct 06 11:25:18 LAPTOP-CKI0PC9R dbus-daemon[133]: [system] Successfully activated service 'org.freedesktop.PackageKit'  
Oct 06 11:25:18 LAPTOP-CKI0PC9R systemd[1]: Started packagekit.service - PackageKit Daemon.  
Oct 06 11:25:18 LAPTOP-CKI0PC9R sudo[1181]: pam_unix(sudo:session): session closed for user root  
Oct 06 11:26:11 LAPTOP-CKI0PC9R kernel: mini_init (119): drop_caches: 1  
Oct 06 11:30:23 LAPTOP-CKI0PC9R PackageKit[1871]: daemon quit  
Oct 06 11:30:23 LAPTOP-CKI0PC9R systemd[1]: packagekit.service: Deactivated successfully.
```

11. Display the SSH service unit file.

Ans)systemctl cat ssh

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ systemctl cat ssh  
# /usr/lib/systemd/system/ssh.service  
[Unit]  
Description=OpenBSD Secure Shell server  
Documentation=man:sshd(8) man:sshd_config(5)  
After=network.target nss-user-lookup.target auditd.service  
ConditionPathExists=!/etc/ssh/sshd_not_to_be_run  
  
[Service]  
EnvironmentFile=-/etc/default/ssh  
ExecStartPre=/usr/sbin/sshd -t  
ExecStart=/usr/sbin/sshd -D $SSHD_OPTS  
ExecReload=/usr/sbin/sshd -t  
ExecReload=/bin/kill -HUP $MAINPID  
KillMode=process  
Restart=on-failure  
RestartPreventExitStatus=255  
Type=notify  
RuntimeDirectory=sshd  
RuntimeDirectoryMode=0755  
  
[Install]  
WantedBy=multi-user.target  
Alias=sshd.service  
sreelekha@LAPTOP-CKI0PC9R:~$
```

12. A website hosted on Nginx is down. Troubleshoot the issue by checking the Nginx service

status, viewing Nginx logs, restarting the Nginx service, and verifying the service status after restart.

Ans) 1)systemctl status nginx
2)journalctl -u nginx
3)sudo systemctl restart nginx
4)systemctl status nginx

If we can see -----Active: active (running)
Nginx is running successfully.

13. Display all service unit files available on the system.

Ans) `systemctl list-unit-files --type=service`

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ systemctl list-unit-files --type=service
```

UNIT FILE	STATE	PRESET
apparmor.service	enabled	enabled
apport-autoreport.service	static	-
apport-coredump-hook@.service	static	-
apport-forward@.service	static	-
apport.service	enabled	enabled
apt-daily-upgrade.service	static	-
apt-daily.service	static	-
apt-news.service	static	-
autovt@.service	alias	-
breakpoint-pre-basic.service	static	-
breakpoint-pre-mount.service	static	-
breakpoint-pre-switch-root.service	static	-
breakpoint-pre-udev.service	static	-
capsule@.service	static	-
chrony-dnssrv@.service	static	-
chrony-wait.service	disabled	enabled
chrony.service	enabled	enabled
chronyd.service	alias	-
cloud-config.service	enabled	enabled
cloud-final.service	enabled	enabled
cloud-init-hotplugd.service	static	-
cloud-init-local.service	enabled	enabled
cloud-init-main.service	enabled	enabled
cloud-init-network.service	enabled	enabled
console-getty.service	masked-runtime	disabled
console-setup.service	enabled	enabled
container-getty@.service	static	-
containerd.service	enabled	enabled

14. Check whether the Nginx service is enabled or disabled.

Ans) `systemctl is-enabled nginx`

```
sreelekha@LAPTOP-CKI0PC9R: ~  
sreelekha@LAPTOP-CKI0PC9R:~$ systemctl is-enabled nginx  
enabled  
sreelekha@LAPTOP-CKI0PC9R:~$
```

15. Explain the difference between restart and reload in Linux service management.

Restart: Stops the service and starts it again. It can temporarily interrupt the service.

`sudo systemctl restart nginx`

Reload: Reloads the service configuration without completely stopping the service. Usually causes little or no interruption.

`sudo systemctl reload nginx`

16. When is daemon-reload used? Demonstrate with the appropriate command.

Ans) It is used when a **systemd unit/service file has been created or modified**, so systemd can reread the unit files.

```
sudo systemctl daemon-reload
```

```
sudo systemctl restart nginx
```

17. Display the current status of all loaded services.

Ans) `systemctl list-units --type=service`

18. View the latest logs related to the Docker service.

Ans) `journalctl -u docker -n 20`

Or

`journalctl -u docker -f`

```
pradeekha@LAPTOP-CKI0PC9R:~$ journalctl -u docker -n 20
Oct 06 11:25:16 LAPTOP-CKI0PC9R systemd[1]: Starting docker.service - Docker Application Container Engine:
827 level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.205190358Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.205399588Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.216630858Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.227698958Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.227698958Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.2282429359Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.259570958Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.282429359Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.511316859Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.517590259Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.518043159Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.544041759Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.551777460Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.551898060Z" level=info Oct 06 11:25:16 LAPTOP-CKI0PC9R systemd[1]: Started docker.service - Docker Application Container Engine.
Oct 06 11:25:16 LAPTOP-CKI0PC9R systemd[1]: Starting docker.service - Docker Application Container Engine:
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.204686358Z" level=info msg="Starting up"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.205190358Z" level=info msg="OTEL tracing is not configured, using no-op tracer provider"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.205399588Z" level=info msg="CDI directory does not exist, skipping: failed to monitor for changes: >"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.205322058Z" level=info msg="CDI directory does not exist, skipping: failed to monitor for changes: >"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.216630858Z" level=info msg="Creating a containerd client" address=/run/containerd/containerd.sock >"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.227698958Z" level=info msg="Loading containers: start."
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.227698958Z" level=info msg="Starting daemon with containerd snapshotter integration enabled"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.228242935Z" level=info msg="Restoring containers: start."
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.259570958Z" level=info msg="Deleting nftables IPv4 rules" errors="exit status 1"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.282429359Z" level=info msg="Deleting nftables IPv6 rules" errors="exit status 1"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.511316859Z" level=info msg="Loading containers: done."
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.517590259Z" level=info msg="Docker daemon" commit=29.1.3-0ubuntu4.1 containerd-snapshotter=true st
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.518043159Z" level=info msg="Initializing buildkit"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.544041759Z" level=info msg="Completed buildkit initialization"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.551777460Z" level=info msg="Daemon has completed initialization"
Oct 06 11:25:16 LAPTOP-CKI0PC9R dockerd[1599]: time="2026-10-06T11:25:16.551898060Z" level=info msg="API listen on /run/docker.sock"
Oct 06 11:25:16 LAPTOP-CKI0PC9R systemd[1]: Started docker.service - Docker Application Container Engine.
```

19. Create a report containing Service Name, Current Status, Enabled/Disabled State, and Recent Log Information.

Ans)

- 1) `systemctl status cron`
- 2) `systemctl is-enabled cron`
- 3) `journalctl -u cron -n 5`

Service Name	Current Status	Enabled/Disabled	Recent Log
Cron	active/running	enabled	Recent Cron logs

20. Perform a complete service health check for SSH and Cron services and document your findings.

Ans)

```
systemctl status ssh
systemctl is-enabled ssh
journalctl -u ssh -n 20
systemctl is-active ssh
```

